PROBLEM STATEMENT:

Bicycle brake pads wear out quietly, and most of us notice only when it's already too risky. This project builds a **smart IoT brake safety system** that keeps an eye on your brake pads and warns you before things get unsafe — so you ride stress-free.

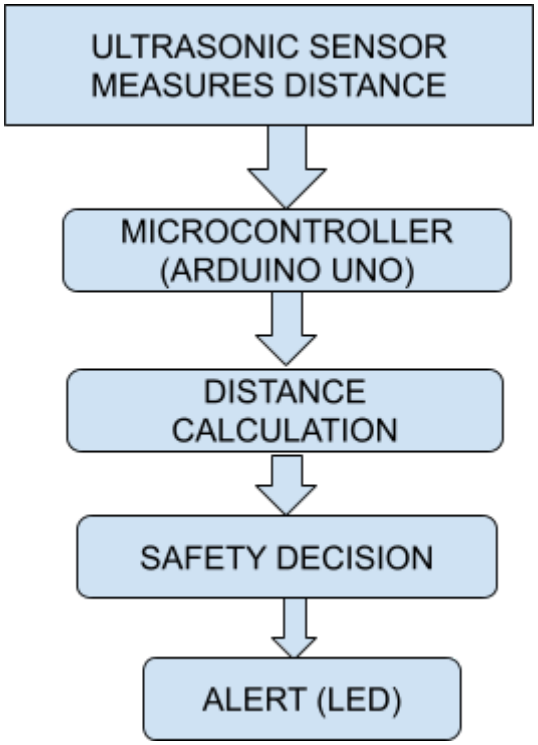
SYSTEM DESIGN:

The system monitors the **condition of bicycle brake pads**.
As brake pads wear out, the **distance between the pad and the rim changes**. This change is used to judge whether the brakes are **safe** or **unsafe**.

· Sensors used

Ultrasonic Sensor (HC-SR04)

· Data flow from sensor to output

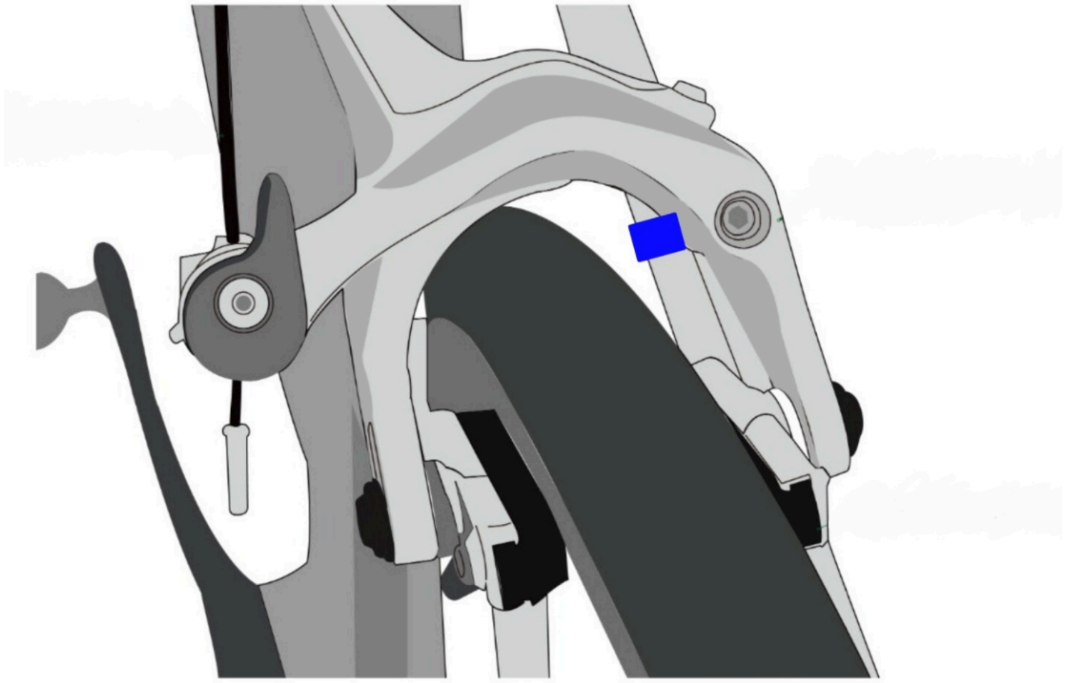


· How the system determines “safe” or “unsafe”

The sensor is placed a the break calliper as shown in the picture,
The sensor points at the break pads,

If the break pad is new, the sensor will sense the break pad because the break pad will obstruct the trigger waves .

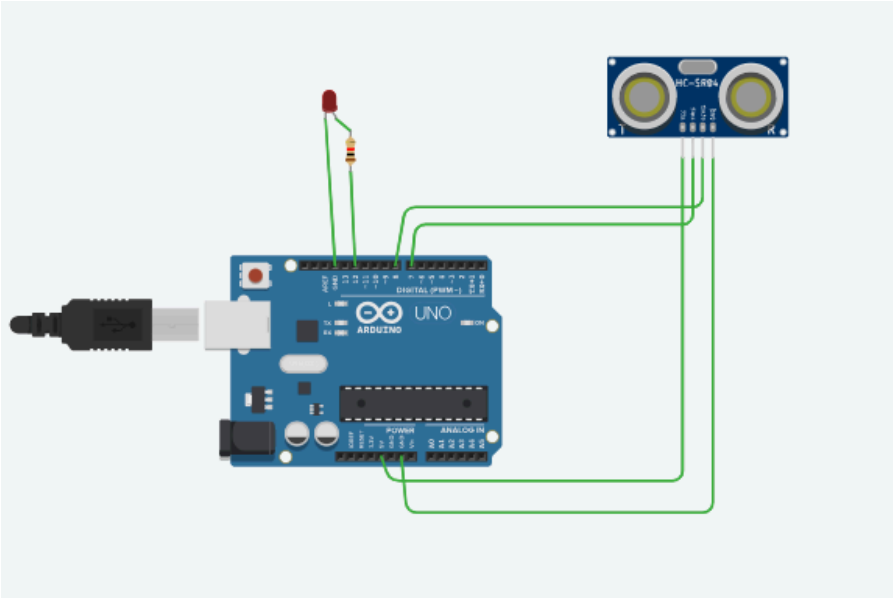
if the break pad wears off, there will be no obstruction in the way of the trigger wave due to the break pad,
Then the sensor will detect some other obstacle in its way, which is definitely at a larger distance than the break.
(Possibly the gear of the tyre, or infinity).
The reading of the sensor is more than the distance (possibly 5 cm),
The led glows giving unsafe feedback.



· Component list

Ultrasonic Sensor (HC-SR04)
Led
Wires
Zip ties and padding (to put the sensor in place)

CIRCUIT DIAGRAM:



Simulation link:

https://www.tinkercad.com/things/jAbavNXjqmu-swanky-jofo-crift/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=xkwuguD-VH-6I_VaX3nAoqs8UnzYtaeRAuHT9D3RjXw